# Shared-memory Parallel Programming with OpenCilk


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# Outline for Today

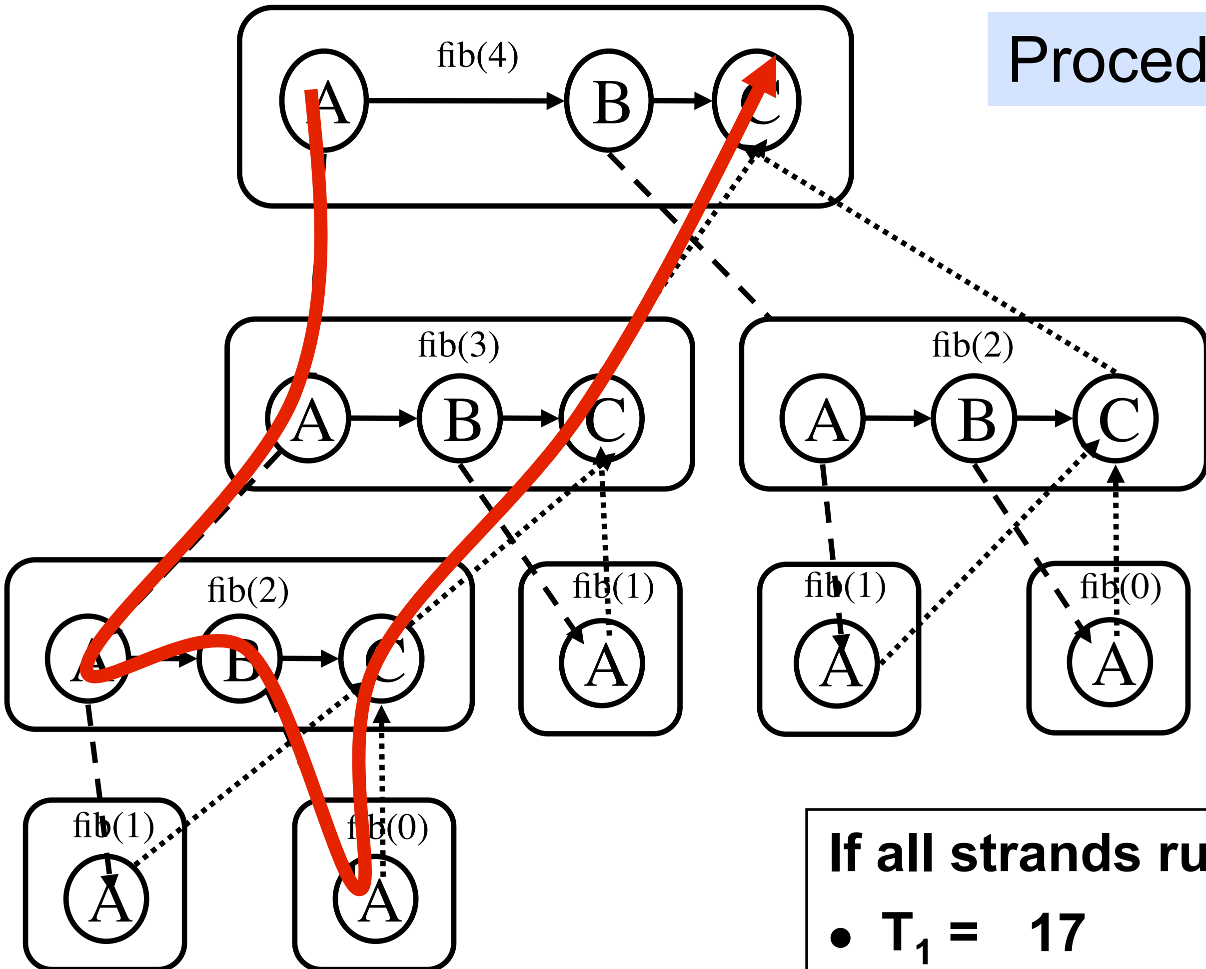
- **Assessing Cilk performance with CilkScale**

# Performance Measures

- $T_s$ = serial execution time
- $T_1$ = execution time on 1 processor (total work), $T_1 \geq T_s$
- $T_p$ = execution time on P processors
- $T_\infty$ = execution time on infinite number of processors
  - longest path in DAG
    - length reflects the cost of computation at nodes along the path
  - known as "critical path length"

# Work and Critical Path Example

Procedure oriented view




**If all strands run in unit time**

- **$T_1$ = 17**
- **$T_\infty$ = 8 (critical path length)**

# CilkScale - Scalability Analysis Tool

- Collects stats of parallel performance
- Measures work, span, parallelism
- Automatically benchmarks program in different number of processors
- Produce tables and graphical plots with the above performance and scalability measurements
- Predicts speedup on various numbers of processors based on work and span

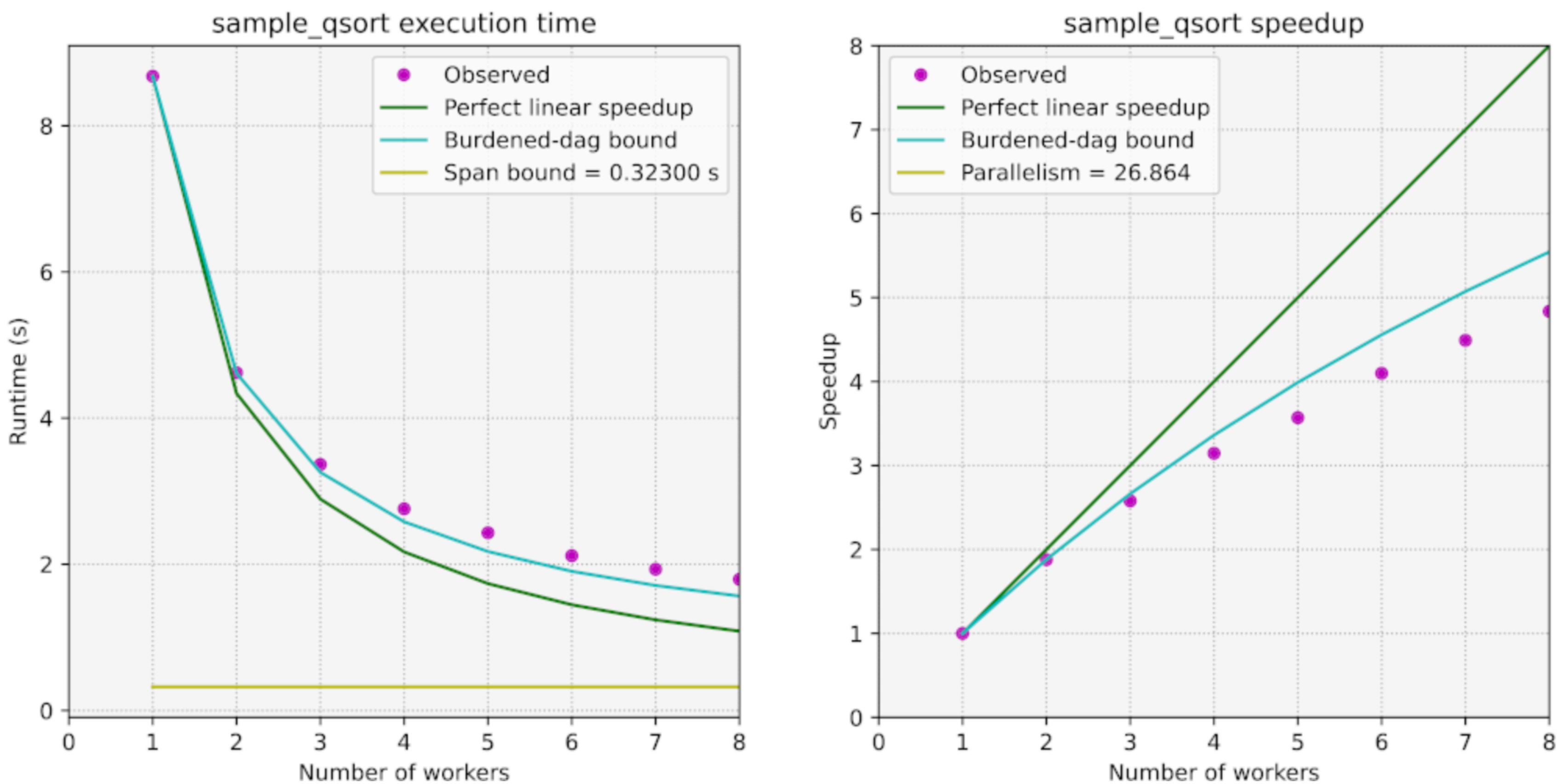
# CilkScale - Scalability Analysis Tool

- **Instruments Cilk program to collect performance measurements during its execution**
- **Cilkscale instrumentation operates in one of two modes**
  - ***Benchmarking* mode: Cilkscale measures the wall-clock execution time of your program**
    - **compile with -fopencilk -fcilktool=cilkscale-benchmark**
  - ***Work/span analysis* mode: Cilkscale measures the *work*, *span*, and *parallelism* of your program**
    - **compile with -fopencilk -fcilktool=cilkscale**
- **By default, Cilkscale only reports performance results for whole-program execution**
  - **fine-grained analyses of specific sub-computations is possible**

# Automatic benchmarks and visualization

ref: https://www.opencilk.org/doc/users-guide/cilkscale/

```
$ python3 /opt/opencilk/share/Cilkscale_vis/cilkscale.py \
    -c ./qsort_cs -b ./qsort_cs_bench \
    -ocsv cstable_qsort.csv -oplot csplots_qsort.pdf \
    --args 100000000
```



# Rule of Thumb

As a rule of thumb, the parallelism of a computation is deemed sufficient if it is about 10x larger (or more) than the number of available processors. On the other hand, if the parallelism is too high — say, several orders of magnitude higher than the number of processors — then the overhead for spawning tasks that are too fine-grained may become substantial

**Burdened Dag Bound:**

Slowed execution if every continuation incurs a steal (incurs overhead of parallelism)

# CilkScale Demo

- **Explore CilkScale for performance analysis using fib example**
  - `cilkscale_run.sh fib 47`
  - `cilkscale_run.sh fib_cutoff 47 2`
  - `cilkscale_run.sh fib_cutoff 47 5`
  - `cilkscale_run.sh fib_cutoff 47 10`
  - `cilkscale_run.sh fib_cutoff 47 20`

# References - I


- **Matteo Frigo, Charles Leiserson, and Keith Randall. The implementation of the Cilk-5 multithreaded language. In PLDI (Montreal, Quebec, Canada, June 17 - 19, 1998), 212-223.**
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- **Guang-Ien Cheng, Mingdong Feng, Charles E. Leiserson, Keith H. Randall, and Andrew F. Stark. 1998. Detecting data races in Cilk programs that use locks. In *Proceedings of the tenth annual ACM symposium on Parallel algorithms and architectures* (SPAA '98). ACM, New York, NY, USA, 298-309.**

# References - II


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- Charles E. Leiserson. Cilk LECTURE 1. Supercomputing Technologies Research Group. Computer Science and Artificial Intelligence Laboratory. http://bit.ly/mit-cilk-lec1
- Matteo Frigo, Pablo Halpern, Charles E. Leiserson, and Stephen Lewin-Berlin. Reducers and other Cilk++ hyperobjects. SPAA '09, 79-90. Talk Slides. April 11, 2009. http://bit.ly/reducers
- Charles Leiserson, Bradley Kuzmaul, Michael Bender, and Hua-wen Jing. MIT 6.895 lecture notes - Theory of Parallel Systems. http://bit.ly/mit-6895-fall03
- Intel Cilk++ Programmer's Guide. Document # 322581-001US.